

Variance : Measures the spread of one variable

$$\text{Var}(X) = E[(X - \mu_x)^2]$$

where $\mu_x = E[X]$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Gramsch



Covariance : Measures how two variables vary together

$$\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Sample Covariance :

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

positive → increase together

Negative → opposite movement

zero → no linear relation

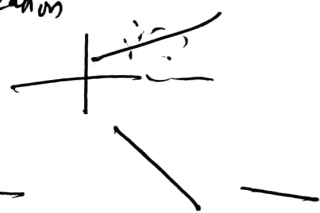
Correlation :- Normalized Covariance

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

where $\sigma_x = \sqrt{\text{Var}(X)}$

$$-1 \leq \rho \leq 1$$

$$\text{sample correlat} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \sqrt{\sum (y_i - \bar{y})^2}}$$



Covariance Matrix :- $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

Covariance matrix $= E[(X - \mu)(X - \mu)^T]$

$$\Sigma_{ij} = \text{Cov}(x_i, x_j)$$

Correlation matrix :-

$$R_{ij} = \frac{\text{Cov}(x_i, x_j)}{\sigma_i \sigma_j} = \begin{bmatrix} \text{Var}(x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Var}(y) \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & \rho_{xy} \\ \rho_{yx} & 1 \end{bmatrix} \quad \frac{\text{Cov}(x, x)}{\sigma_x \sigma_x}$$

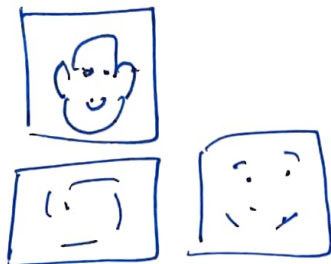
⇒ This significantly reduces the storage cost without much ↓ quality

Data Compression → storing relevant information

10k dimension

multiple images of the same person → lighting, contrast, ...
for for in the original space to closer in the new space.

We will important characteristics of the new data.



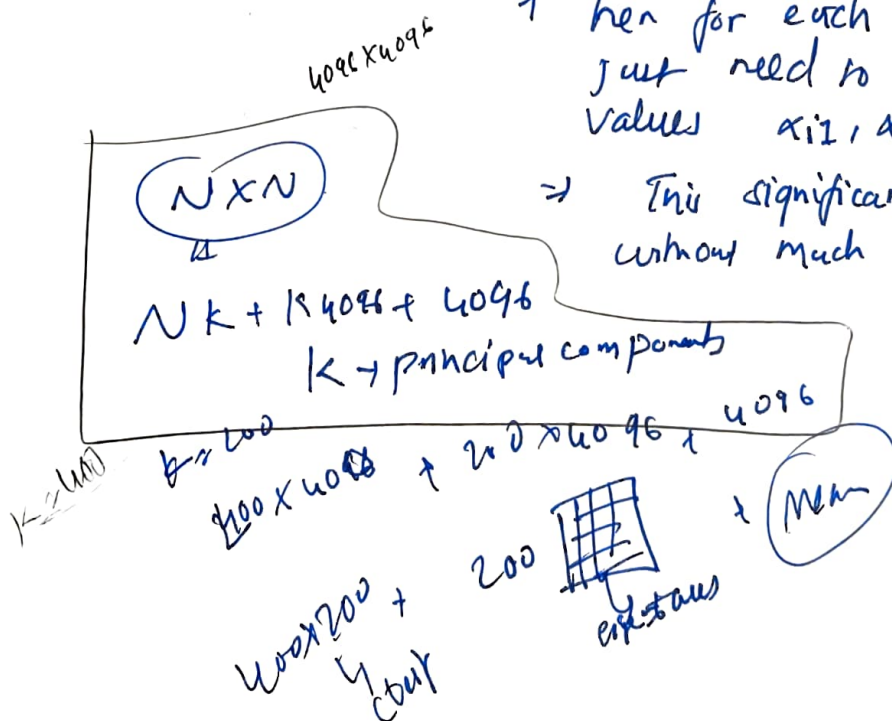
⇒ These images are called eigenfaces and form a basis for representing any face in our database.

⇒ In other words, we can now represent a given image (face) as a linear combination of these eigenfaces

⇒ In practice, we just need to store $P_1, P_2, \dots, P_K$ (one time storage)

⇒ Then for each image i we just need to store the scalar values $\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iK}$

⇒ This significantly reduces storage without much ↓ quality.



Lec. 68 Singular Value decomposition

Let us get more perspective on before moving ahead
Let $v_1, v_2, \dots, v_n$ be the eigen vectors of A and let
 $d_1, d_2, \dots, d_n$ be corresponding eigen values.

$$Av_1 = d_1 v_1, Av_2 = \dots = Av_n = d_n v_n$$

if a vector x in $\mathbb{R}^n$ are represented using $v_1, v_2, \dots, v_n$
as basis then

$$x = \sum_{i=1}^n \alpha_i v_i$$

$$Ax = \sum_{i=1}^n \alpha_i Av_i = \sum_{i=1}^n \alpha_i d_i v_i$$

• The matrix multiplication reduces to a scalar multiplication
if the eigen vectors of A are used as a basis.

• so for all the discussion was centered around
square matrices $A \in \mathbb{R}^{n \times n}$

• What about rectangular matrices $A \in \mathbb{R}^{m \times n}$?
can they have eigen vectors?

• Is it possible to have $A^{m \times n} x^{n \times 1} = x^{n \times 1}$?
 $A^{n \times n} x^{n \times 1} = x^{n \times 1}$ Not possible! not $n \times 1$ $n \times 1$!

→ The result of $A^{m \times n} x^{n \times 1}$ is a vector belonging to
 $\mathbb{R}^m$ (where as $x \in \mathbb{R}^n$)

→ do we do miss out an advantage that a basis
of eigen vectors provides for square matrices
(i.e., converting matrix multiplication into scalar
multiplication?)

→ we will see the answer to this question over
next few slides

- Note that matrix A maps provides a transformation

$$\mathbb{R}^n \rightarrow \mathbb{R}^m$$

- What if we could have pair of vectors $(u_1, v_1), (u_2, v_2), \dots, (u_k, v_k)$ such that $v_i \in \mathbb{R}^n$, $u_i \in \mathbb{R}^m$ and $(Av_i = \sigma_i u_i)$

$v_1, \dots, v_k, \dots, v_n$ are orthogonal & thus form a basis.

$u_1, \dots, u_k, \dots, u_m$ are orthogonal & thus form a basis

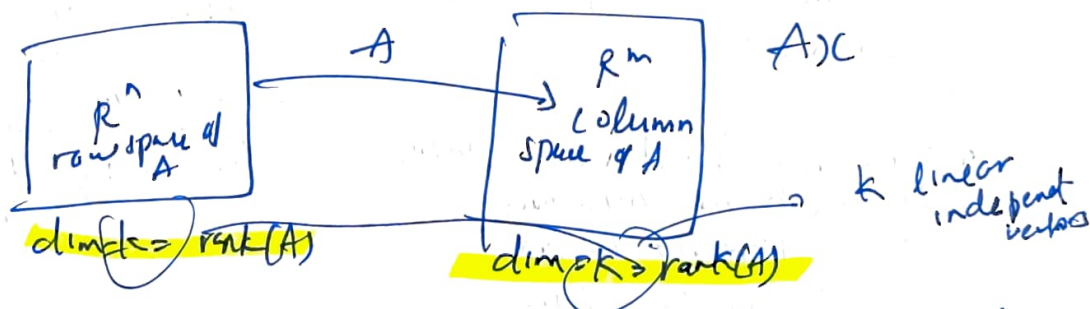
$\Rightarrow$ Now if every vector $x \in \mathbb{R}^n$ is represented using the basis

$$\Rightarrow \underline{x} = \sum_{i=1}^n x_i v_i \quad \text{[not we are using } k \text{ instead of } n \text{ will see further]}$$

$$\begin{aligned} Ax &= \sum_{i=1}^n x_i Av_i \\ &= \sum_{i=1}^n x_i \sigma_i u_i \\ &= \sum_{i=1}^k x_i \sigma_i u_i \end{aligned}$$

⊙

$\Rightarrow$ Once again the matrix multiplication reduces to scalar multiplication



- $\mathbb{R}^n$ - space of all vectors which can multiply with A to give Ax [this is the space of input of the function]
- $\mathbb{R}^m$ - space of all vectors which are output of the function Ax
- We are interested in finding a basis U, V such that
 - V - basis for input
 - U - basis for output
- such that if the inputs and outputs are represented using basis then the operation Ax reduces to a scalar operation

Rank $A = 1$ means

The rank of a matrix = number of linearly independent columns / rows
 $\rightarrow$ dimension of column space
 $\text{rank}(A) = r$

Nullity = dimension of Null space
 How many free directions exist in solution of $AX=0$

Null space: = all vectors satisfying $AX=0$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$x_1 + x_2 = 0$$

$$x = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

exam $\rightarrow$ Null space has one basis vector
 $\text{nullity}(A) = 1$

Rank-Nullity theorem:

$$\text{rank}(A) + \text{nullity}(A) = n$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \text{ rank} = 1$$

$$\text{nullity} = 2$$

$n = 1$ or 0 of columns

Image / Range

Column space: Span of columns

Row space: Span of rows

Basis: Set of independent vectors

Dimensions: Number of basis vectors

Linear independence: No vectors redundant

Pivot columns: Determine Rank

Free variables: Determine nullity

Kernel: Another name of null space

Imagine A stretches space

You are searching for a scaling factor λ , that when removed from A leaves some direction completely flattened.

$$(A - \lambda I) v = 0$$

means:

In direction v , A and λ times Identity do exactly the same thing so their difference is zero.

That direction v is the eigenvector

find the values of λ for which $A - \lambda I$ loses rank and therefore has a non zero direction v that gets mapped to zero

λ 's are eigenvalues and those special directions have no eigenvectors.

Determinant

$\rightarrow$ whether matrix is invertible

> 0 $\rightarrow$ scales volume, orientation preserved

< 0 $\rightarrow$ scales volume, orientation flipped

$= 0$ $\rightarrow$ space collapsed

$|\det A|$ = how much area/volume change

Sign of $\det(A)$

positive $\rightarrow$ no flip

negative $\rightarrow$ flip (mirror/reflection)

how much a space changes. How much, whether transformation turns space inside out

Product of eigen values

$$= \det(A)$$

Eigenvectors show the special

Eigenvalues show the special direction of transformation

> 1 $\rightarrow$ stretch

0 to 1 compressed

negative $\rightarrow$ flipped

0 $\rightarrow$ completely destroyed

Gram matrix:-

$$\text{cov} \Rightarrow \Sigma = \frac{1}{m} X_c^T X_c$$

$$\left(\frac{1}{m} X_c^T X_c \right) v = d v$$

$$\boxed{\Sigma p_i = d_i p_i}$$

$$\text{Gram matrix} \Rightarrow \frac{1}{m} X_c X_c^T \Rightarrow \lambda' v'$$

$$\boxed{G v'_i = \lambda'_i v'_i}$$

$$\frac{1}{m} X_c X_c^T v'_i = \lambda'_i v'_i$$

$$X_c^T \left(\frac{1}{m} X_c X_c^T v'_i \right) = X_c^T (\lambda'_i v'_i)$$

$$\Sigma (X_c^T v'_i) = \lambda'_i (X_c^T v'_i)$$

$$\boxed{\Sigma p_i = \lambda p_i}$$

$$p_i = X_c^T v'_i$$

$$\text{A.A } X_c^T v'_i$$

$$\text{need to normalize} \Rightarrow \|X_c^T v'_i\|^2 = (X_c^T v'_i)^T (X_c^T v'_i)$$

$$\Rightarrow v'_i{}^T X_c X_c^T v'_i$$

$$\Rightarrow v'_i{}^T \underbrace{m G v'_i}_{\text{circled}}$$

$$\Rightarrow m \lambda v'_i{}^T v'_i$$

$$\Rightarrow m \lambda_i$$

$$m G \Rightarrow$$

$$G v'_i = \lambda'_i v'_i$$

$$10 \times 100 \times 4096$$

$$\|X_c^T v'_i\| = \sqrt{m \lambda_i}$$

$$p_i \Rightarrow \frac{X_c^T v'_i}{\sqrt{m \lambda_i}}$$

$\Rightarrow$

$$10 \times 100 \times 4096$$

- ⇒ what do you mean by saying that dimension of row space is k ?
 $x \in \mathbb{R}^n$ then why is the dimension not n .
 ⇒ It means that of all possible vectors in $\mathbb{R}^n$ only a subspace of vectors lying in $\mathbb{R}^k$ can act as inputs to Ax and produce a non zero output. The remaining vectors in $\mathbb{R}^n$ will produce a zero output.

then we need only k dimensions to represent x

$$x = \sum_{i=1}^k \alpha_i \cdot v_i$$

- Let look at a way of writing this matrix operation

$$Av_1 = \sigma_1 u_1, Av_2 = \sigma_2 u_2, \dots, Av_k = \sigma_k u_k$$

$$\boxed{A_{n \times n} V_{n \times k} = U_{n \times k} \Sigma_{k \times k}}$$

diagonal matrix

before EVD:

$$Av = u \Sigma$$

$$Ax = Ax$$

$$\text{here } Av = u \Sigma$$

$$A^T v = u \Sigma$$

- if we have k orthogonal vectors $V_{n \times k}$ then using gram schmidt orthogonalisation, we can find $n-k$ more orthogonal vectors to complete the basis of $\mathbb{R}^n$

[we can do the same for u]

$$A_{m \times n} V_{n \times n} = U_{m \times n} \Sigma_{m \times n}$$

$$U^T A V = \Sigma \quad [U^T = U^T] \quad A = V \Sigma V^T \quad V^T = V^T$$

- Suppose U, V and Σ exist then

$$A^T A = (U^T \Sigma V^T)^T (U^T \Sigma V^T)$$

$$= \cancel{V \Sigma^T U^T} U^T \Sigma^T U \Sigma V^T$$

$$\Rightarrow V \Sigma^T U^T U \Sigma V^T$$

$$A^T A = V \Sigma^T \Sigma V^T$$

↳ since orthogonal

→ what does this look like? Eigen value decomposition of $A^T A$

• Similarly we can show that

$$A A^T = U \Sigma^2 U^T \quad \left| \quad U \Sigma^2 U^T \text{ do proper matrx}$$

SVD all conditions for a matrix?

→ first square matrix

Under SVD → Rectangular Matrix
 ↳ always exist Any matrix

$$[A]_{m \times n} = \begin{bmatrix} \uparrow & & \uparrow \\ u_1 & \dots & u_k \\ \downarrow & & \downarrow \end{bmatrix}_{m \times k} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_k \\ & & & \dots \end{bmatrix}_{k \times k} \begin{bmatrix} \sigma_1 & \dots & \sigma_k \\ & \ddots & \\ & & \sigma_k \\ & & & \dots \end{bmatrix}_{k \times n}$$

$$\Rightarrow \sum_{i=1}^k (\sigma_i u_i v_i^T)$$

↳ Smallest singular value

computation

$$m \times n \Rightarrow \begin{matrix} u_i & v_i \\ (m + n + 1) \end{matrix} k$$

all this Reconstruction

most important are decided by singular values

Theorem: $\sigma_1 u_1 v_1^T$ is the best rank-1 approximation of the matrix A . $\sum_{i=1}^2 \sigma_i u_i v_i^T$ is the best rank-2 approximation of a matrix A . In the other words, the solution to

$$\min \|A - B\|_F^2 \text{ is given by}$$

$$B = \sum_{i=1}^k \sigma_i u_i v_i^T \text{ (minimizing reconstruction error of } A \text{)}$$

$$\sigma_i = \sqrt{\lambda_i}$$

$$\hat{A} \Rightarrow \min \|A \circ B\|_F^2$$

Frobenius norm

optimization problem